

Minor Exam

Course Code: CYL1010 (Chemistry) Exam Venue: LHB 110 Time: 2PM-4PM

Instructors: K. Mitra and S. Pal Date: 19th Feb 2026 Total Marks = 60

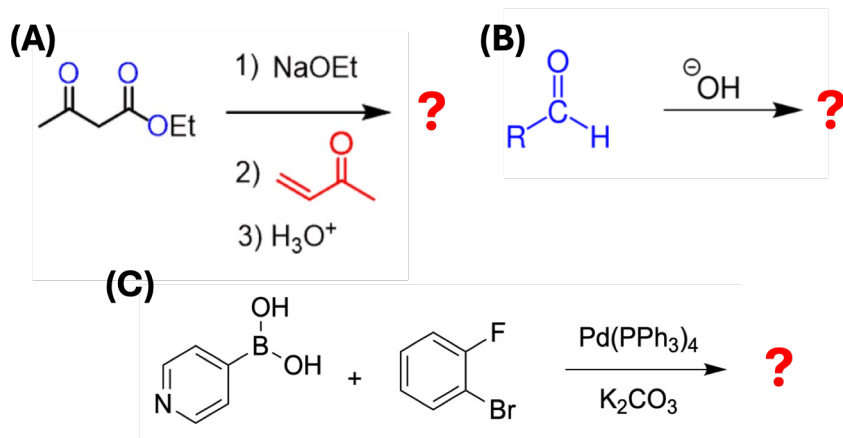
PART A – Organic Chemistry and Transition Metal Chemistry [40 Marks]

Q1. Calculate the **octahedral crystal field splitting stabilization energy (CFSE)** in kJ/mol for $[\text{Fe}(\text{CN})_6]^{4-}$, if the wavelength of the most intensely absorbed light is 305 nm. Include the octahedral crystal field splitting diagram and configuration for this complex with proper labelling of the names of the d-orbitals, and the appropriate orbital sets. [Planck's constant = 6.6261×10^{-34} Js; Speed of light = 2.9979×10^8 m/s; Atomic number of Fe = 26]. [2 Marks]

Q2. Estimate the **pH and pOH** of a 0.03091 M solution of hydroiodic acid ($K_a = 10^9$). [1 Mark]

Q3. Draw the **major product** obtained from aldol condensation of acetone and propanal carried out in acidic medium with heating. Draw the **step-by-step reaction mechanism** showing proper arrow movements for each step. Mark all the **non-equivalent hydrogens** in the product and arrange them in **increasing order of their acidity**. Explain your answer by detailing the factors that affected the acidity order. [1+2+2 = 5 Marks]

Q4. Predict only the **Major product(s)** for the following reactions: [1x3 = 3 Marks]



Q5. Draw the structure of **Wilkinson's catalyst** and give **one example of a reaction** in which it acts as a catalyst. [0.5+0.5 = 1 Mark]

Q6. Calculate the **Octahedral Crystal Field Stabilization Energies (CFSE, Δ_o)** for the following complexes: [0.5 x 4 = 2 Marks]

(A) $[\text{NiCl}_4]^{2-}$ (B) $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$ (C) $[\text{Cu}(\text{NH}_3)_6]^+$ (D) $[\text{Ru}(\text{bpy})_3]^{3+}$

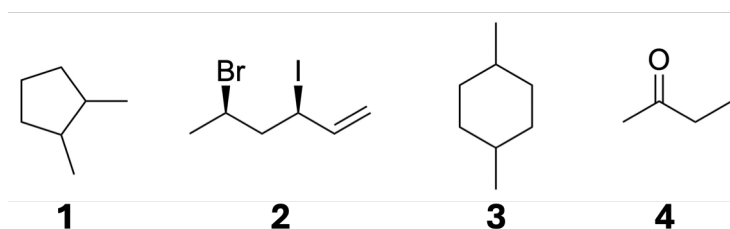
[Atomic numbers of Ni, Mn, Cu and Ru are 28, 25, 29 and 44 respectively. Mention **electronic configuration and crystal field orbital splitting diagram** with proper labels]

Q7. Assign the colors from column B to the complexes in column A based on **crystal field theory (CFT) and d-d orbital splitting**. Consider that only d-d transition contributes to the color of the complex. **Explain your answer in detail** including CFT configuration, diagram with proper labelling and factors contributing towards observed color of the complexes. Atomic number of Co is 27. [3 Marks]

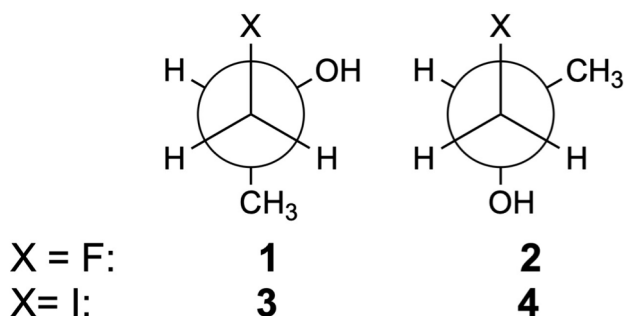
Complex (Column A)	Color (Column B)
$[\text{Co}(\text{NH}_3)_6]^{2+}$	Yellow
$[\text{Co}(\text{H}_2\text{O})_6]^{2+}$	Pink
$[\text{CoCl}_4]^{2-}$	Blue

Q8. Explain how **photoactive molecules/ photosensitizers** can be used to kill cancer cells selectively in anticancer modalities such as **Photodynamic Therapy (PDT) and Photouncaging**. [2 Marks]

Q9. Justify whether the below molecules are **chiral or achiral or can be both**. Explain by **drawing the stereoisomers**, noting the **symmetry elements** and **chiral centers** present in the molecules. [1 x 4 = 4 Marks]



Q10. Arrange the following conformers **1-4** of 2-Haloethanols (as shown below) in **increasing order** of their relative stabilities. Explain your answer clearly considering the strains and other interactions. [2 Marks]



Q11. Name **two human metalloproteins/metalloenzymes** which contains the following transition metals (**A**) Fe, (**B**) Cu, (**C**), Zn and (**D**) Ni. [1x4 = 4 Marks]

Q12. What is **vulcanization** and explain how it affects the **structure and properties** of crude rubber? [2 Marks]

Q13. Draw the **exact chemical structure of the active site of Hemoglobin** (including any interactions with the protein side chains) and state its **function**. State **two diseases** associated with Hemoglobin malfunctioning. [2+1 =3 Marks]

Q14. Compare the **extinction coefficient (ϵ)** values of the complexes: $[\text{MnCl}_4]^-$, $[\text{Co}(\text{CN})_6]^{4-}$, and $[\text{FeF}_6]^{3-}$ based on Crystal Field Theory and considering only d-d transitions. **Explain your answer in detail** including CFT configurations, diagram with proper labelling and factors deciding the ϵ values. [3 Marks]

Q15. The R enantiomer of the compound carvone (chiral) has an optical rotation of 125° . If a pure sample of synthesized carvone has an optical rotation of 100° , what is **enantiomeric excess** of the sample? [Show how you deduced the answer] [2 Marks]

-----END of PART A Questions--- PTO for PART B Questions -----

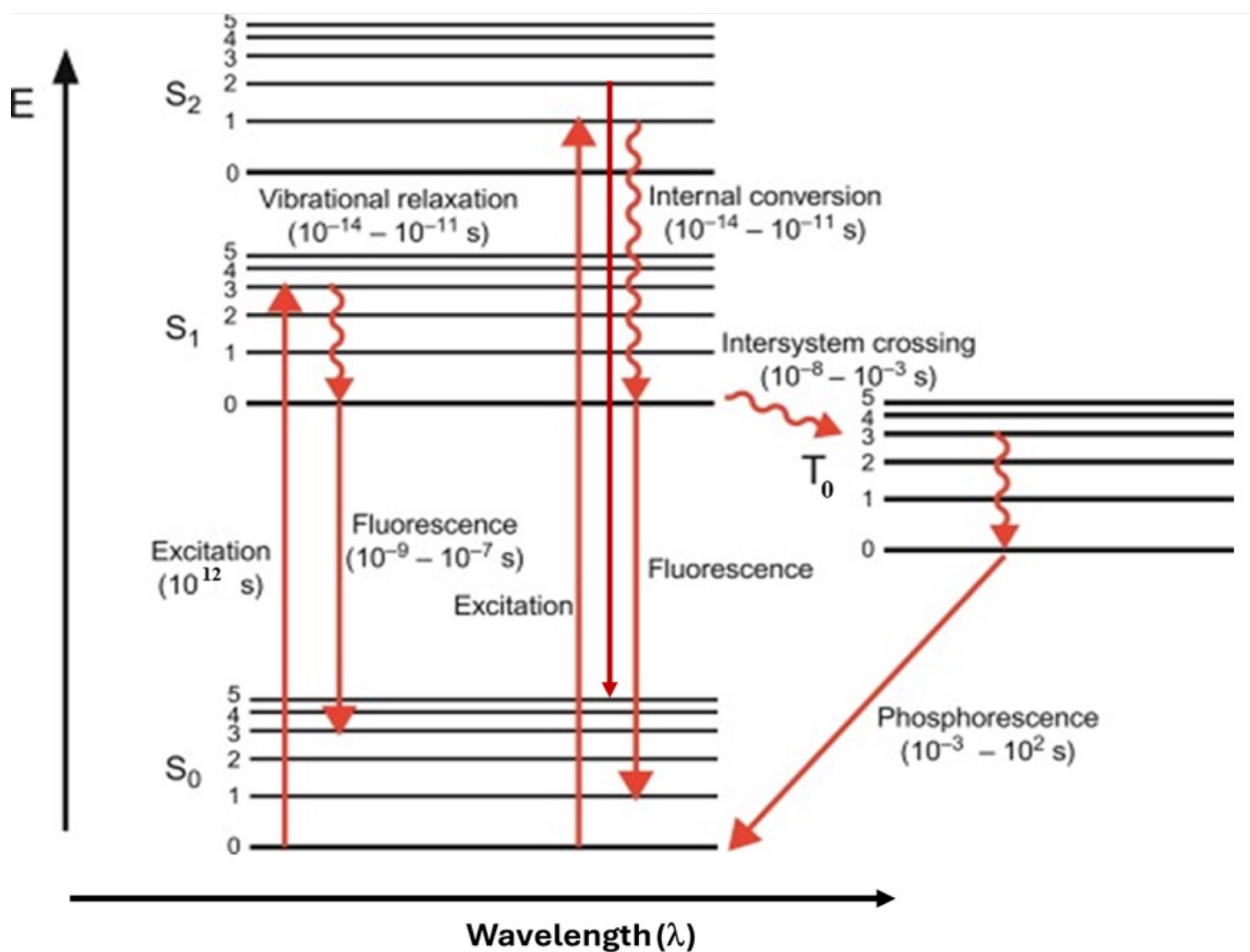
PART B – Basic Spectroscopy [20 Marks]

All questions are mandatory. Answer without justification will fetch no marks.

Important constants: $h = 6.626 \times 10^{-34} \text{ Js}$; $c = 3.8 \times 10^8 \text{ ms}^{-1}$; $N_A = 6.314 \times 10^{23} \text{ mol}^{-1}$

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Q16. What is the name of the following diagram? Name the various processes it represents. Identify at least three misrepresentations in the diagram. Please note all bold arrows represent photon emission or absorption. [0.5+1+1.5 =3 Marks]



Q17. Select the **correct option with one line justification**. Show calculations wherever possible. [1x5 = 5 Marks]

Q17a) While recording fluorescence if the excitation wavelength is changed but still overlaps with the absorption band, the emission spectrum will:

- A. Shift to a different wavelength
- B. Remain essentially unchanged
- C. Disappear
- D. Become phosphorescent

Q17b) Consider a flexible molecule that has two photophysical processes radiative due to fluorescence and non-radiative process due to internal conversion. The molecule has fluorescence quantum yield $\Phi = 0.25$ and excited state lifetime $\tau = 5 \text{ ns}$. On rigidification the rate of internal conversion reduces to half. What will be the new quantum yield?

- A. 0.40
- B. 0.50
- C. 0.67
- D. 0.75

Q17c) Transmittance of a 0.1 M stock solution of molecule A is 60% and that of 0.1 M of molecule B is 50% at a given wavelength. On mixing, both A and B are diluted to half of their stock solution concentration. What will be the transmittance of the mixture?

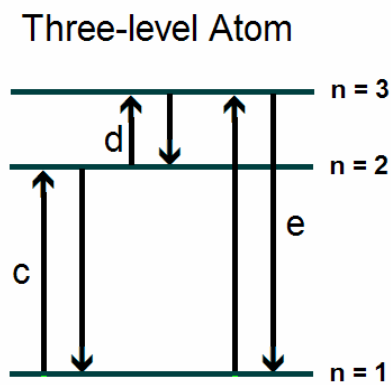
- A. 60%
- B. 50%
- C. 45%
- D. 54%

Q17d) Two molecules A and B have same k_r , but molecule A has larger quantum yield Φ . This means:

- A. A has larger absorption coefficient
- B. A has smaller k_{nr}
- C. A has larger Stokes shift
- D. A has longer excitation wavelength

Q17e) In the diagram below, certain electronic transitions are shown. If the transition marked as 'c' in the figure is associated with absorption or emission of a red photon, what colour photon would be absorbed in making the transition marked 'd'.

- A. Blue
- B. Ultraviolet
- C. Infrared
- D. Green



Q18. Consider a molecular excited state with lifetime of 1 nanosecond. Following absorption there also is intersystem crossing at a rate of 10^3 per second and internal conversion at a rate of 10^6 per second.

a) What are the **quantum yields** for each of these processes? [2 Marks]

b) What is the **natural line width** in cm^{-1} . [1 Mark]

Q19. 1 gram of a compound with a molecular mass of 100 g/mol is dissolved in 1 litre of water; the solution is used to fill a 1.00 cm cuvette. The molar extinction coefficient for this compound is $\epsilon_{340} = 100 \text{ M}^{-1}\text{cm}^{-1}$. Calculate the **absorbance**. [2 Marks]

Q20. Atomic emission spectrum consists of lines while molecular absorption spectrum consists of bands. **Give reasons**. [2 Marks]

Q21. You wish to carry out a photochemical reaction using a UV lamp filtered to generate 290 nm radiation with 25 W power. Your reactants are sufficiently concentrated to absorb all the radiation from this lamp, and the reaction's photochemical product (overall) quantum yield is 0.28. How **many moles of product** will you generate per hour of illumination? [5 Marks]

-----**END of ALL Questions**-----